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Advancing real-time plant disease detection: A lightweight deep learning approach and novel dataset for pigeon pea crop

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ABSTRACT

Plant disease detection and early disease treatment are essential for sustainable crop production. Computer vision for crop science is overgrowing with the advancement in deep learning. Real time plant disease detection poses a challenge due to the unpredictable spread of diseases within the plant, environmental factors, and the scarcity of real field datasets. The proposed work systematically addresses these issues through three key components: (a) Collaboratively generating the novel pigeon pea image dataset from agricultural fields, in partnership with 20 Agricultural Research Centers (ARS) and governmental agencies spanning 18 Indian states. (b) The design of lightweight and high-performance models for real-time plant disease detection in resource-constrained devices. (c) The extraction of multiscale feature of plant diseases using Multi-kernel Depthwise separable Convolutions. The proposed lightweight Lite-MDC architecture uses the Multi-kernel Depthwise separable Convolutions (MDsConv). The MDsConv module captures spatial features across various scales while maintaining a lightweight design. It effectively extracts multi-scale information to characterize plant diseases, accommodating their diverse scale. Proposed architectural approach significantly reduces computational complexity, employing only 2.2 million parameters, which is a 62% reduction compared to the standard VGG16 architecture. The proposed method outperforms the state-of-the-art networks such as InceptionV3, VGG16, ResNet50, DenseNet, MobileNet, MobileNetV3, NASNet, and EfficieNetB0 on the proposed pigeon pea dataset with 94.14% accuracy. Notably, the method achieves a 34 Frames Per Second (FPS) inference on an NVIDIA P100 GPU. Furthermore, its performance is validated across publicly available datasets, including the plant village dataset, Cassava, and apple leaf datasets, yielding 99.78%, 86.4%, and 97.2% accuracy, respectively. The Lite-MDC model exhibits the potential for real-time plant disease detection on resource-constrained edge devices such as Agriculture robots and drones.

1. Introduction

Globally, pigeon pea is the sixth most significant legume in Asian countries [1]. Pigeon pea is a rich source of protein, high in fiber, and contains essential amino acids. However, the numerous occurrences of plant diseases may significantly restrict the plants' health and affect the yield. Annually, plant diseases cause production loss between 12-23 million dollars [2] [3]. Presently, the detection of numerous plant diseases predominantly relies on human visual assessment. Farmers and experts depend on visually inspecting plants to spot diseases. This method has limitations. Subjective interpretations and visual fatigue can result in inaccurate assessments [4]. When diseases are misdiagnosed, it can disrupt the monitoring and control procedures, resulting in the improper application of pesticides, diminished plant quality, and reduced yields [5]. Considering the potential repercussions of misidentifying plant diseases, it becomes evident that precise disease identification is indispensable for preserving plant health and attaining optimal yields.

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Fig. 1. Challenges for pigeon pea disease detection in Pigeon pea dataset.

Deep learning serves as an advanced tool in the realm of plant disease detection, enabling the automated identification of disease-related features [6]. However it comes with its own set of challenges like (a) The availability of real field datasets. (b) A Randomly distributed plant diseases on leaves, stems, and roots, and their diversity in texture, size, and shape. (c) Environmental challenges include image blur due to wind, leaf overlapping, complex background, and viewpoints, as shown in Fig. 1. (d) The development of lightweight yet high-performance models tailored for real-time plant disease detection on resource-constrained embedded devices poses another significant challenge. Convolutional neural networks (CNNs) have emerged as a promising solution to tackle the challenges of disease identification and classification. CNN architectures are specifically designed for image analysis and can help to overcome the limitations of traditional visual inspection methods. However, these architectures are computationally expensive. For example, Standard VGG16 architecture has 92 layers and requires 140 M parameters for implementation [7], [8]. To address this limitation, we proposed Lite-MDC architecture with multi-kernel depthwise One of the advantages of proposed architecture is its ability to maintain state-of-the-art performance while reducing computational complexity. This technique optimizes the model's efficiency without sacrificing accuracy. This paper also proposes the First Real field pigeon pea image dataset. The key contributions of the proposed research work are summarized as follows:

1. Proposed the first novel pigeon-pea disease image dataset includes high resolution images collected from 18 geographical locations across India.
2. Proposed MDsConv block with 3×3 and 5×5 helps to extract the multiscale feature of plant diseases.
3. Proposed a Light weight architecture Lite-MDC for pigeon pea disease detection reduces computational complexity and shows state-of-the-art results with only 2.2 M parameters with 34 FPS suitable for real time application.

To evaluate the effectiveness of the proposed approach, extensive experiments were conducted to compare against public datasets and existing state-of-the-art methods such as VGG-16 [9], Xceptions [10], NasNet mobile [11], MobileNet [12], MobileNetV2 [13], ResNet50 [14], InceptionV3 [15] and DenseNet121 [16] and EfficientNetB0 [17]. This rigorous testing provides a robust evaluation of the model's performance and highlights its potential advantages over other approaches.

The rest of this paper is structured in the following manner. Section 2 reviews the literature work. Section 3 proposes the pigeonpea dataset construction details; Section 4 introduces the proposed method for plant disease detection. Section 5 enumerates the experimental details, Section 6 illustrates the results and analysis with discussion. Finally, Section 7 concludes the work with mention of some future direction.

2. Related work

In recent years, the agricultural sector has witnessed a growing interest in the detection and management of plant diseases, driven by image processing techniques and advancements in machine learning techniques.

2.1. Image processing and machine learning for plant diseases detection

Image processing and machine learning techniques are extensively applied for the detection of plant diseases in a variety of crops, including tomatoes, potatoes, sugarcane, maize, and numerous others [18], [19], [20]. These methods encompass a range of image processing-based approaches, such as hand-engineered features, scale-invariant feature transforms, histogram-based methods, thresholding techniques, and segmentation methods [21], [22], [23], [24]. Hlaing and Zaw [25] employ statistical texture and color features to create a modified representation of the Scale Invariant Feature Transform (SIFT) for the classification of tomato plant diseases. The authors enhance the complexity of the SIFT representation by introducing novel statistical features while reducing the dimensionality of the data. Chaudhary et al. [21], proposed color transform based approach for disease spot detection which includes CIELAB color model and binary thresholding for disease spot segmentation. This approach is fail to identify diseased spot having similar color like leaf vein. In [22], the author has introduced a method for severity estimation of sugarcane diseases, which utilize the triangle threshold methods for the precise segmentation of disease lesions. This method is not suitable for accurate segmentation of disease lesion due to varying characteristics of disease. As machine learning continues to progress in diverse domains, scientists and researchers have harnessed these advancements to employ machine learning techniques for the identification of plant diseases [26] [18] [27] [28]. Al-Hiary et al. [26] uses Artificial Neural Networks (ANN) and K-means clustering to classify plant diseases. Proposed approach fails to work with real field images. Sankaran and Ehsani [18] The author applies visible-near-infrared and mid-infrared spectroscopy to classify leaves infected with huanglongbing and citrus canker. This classification process involves the use of quadratic discriminant analysis (QDA) and k-nearest neighbor (kNN) algorithms. KNN classifier is computationally expensive and demands storage of entire training dataset in memory especially when dealing with large dataset. Furthermore, Revathi et al. [19] extracted skew divergence color variance, edges, and texture features of diseased leaf and uses deep forward neural network for classification of cotton leaf spot disease. This approach is only suitable for controlled environmental setting. Karmokar et al. [20] propose a Neural Network Ensemble (NNE) method with a combination of features for identifying tea leaf diseases. Proposed approach is sensitive to variation in tea leaf diseases. In [27] The author has introduced a method for detecting grape diseases by analyzing leaf texture and employing pattern recognition techniques. The extracted texture patterns

are categorized using a multiclass Support Vector Machine (SVM) classifier. However, SVM may exhibit reduced performance when applied to datasets with substantial noise. Machine learning classifiers have been noted to rely on manual feature extraction and are highly dependent on the available data. However, with the superiority of deep learning techniques over traditional methods, researchers have started utilizing deep learning architectures for detecting plant diseases.

2.2. Deep learning for plant diseases detection

The progress made in deep learning techniques for image classification, segmentation, and object detection is transforming the respective field significantly. Moreover, deep learning is widely used in the agro-industry for crop monitoring, diseases detection, disease severity estimation, yield estimation, etc. [6], [5]. Mohanty et al. [5] used a plant village dataset and transfer learning approach with deep CNN architectures AlexNet and GoogLeNet to recognize plant species and diseases. However AlexNet and GoogLeNet are computationally heavy architectures and not suitable for edge device applications. Kawasaki et al. [29] uses deep CNN with weight filter sizes of 12×12 and 5×5 to distinguish between disease-infected and healthy leaves from the cucumber leaf dataset. This study is limited to single source dataset and does not consider the effects of sunlight, drooping and various real field conditions. Sladojevic et al. [30] use CaffeNet, an advanced version of AlexNet with a deep CNN architecture, to classify plant diseases with images available on the internet as a dataset. This approach achieves an impressive accuracy rate of 96.3% across 13 distinct plant disease categories. Zhang et al. [31] use the transfer learning approach using AlexNet, GoogleNet, and ResNet deep CNNs with a maximum accuracy of 97.28% for identification of tomato leaf disease. Further, researchers focus on disease classification as well as disease localization. Selvaraj et al. [32] use object detection deep CNN architecture single shot detector (SSD) for banana disease detection. Study uses MobileNetV1 architecture as backbone for banana disease feature extraction. In [33], the author uses YOLOV3 architecture as the baseline for detecting tomato diseases and pests from the tomato dataset. YOLOV3 architecture having 64 M parameters which limits its real-time application on resource constrained devices. The study by Wang et al. [34] uses the Plant Village dataset to train and evaluate a group of CNNs, to assess apple black rot disease severity. In this study fine-tuned VGG16 achieve 90.4% accuracy with 140 M parameter. The authors in [35], use the VGG19 architecture to identify disease classifications such as early blight, late blight, and healthy leave in the potato leaves Dataset, with an accuracy of 97.8%. In [36], proposes a Deep CNN architecture for classifying ten different types of rice leaf disease from rice leaf datasets containing healthy and infected images. While deep learning architectures have shown to be effective at detecting plant diseases, their computational and memory requirements make them un-suitable for real-world applications. This problem can be overcome by designing efficient CNN models.

2.3. Efficient deep neural networks

Real-time applications on resource-constrained devices demand lightweight CNN architectures [37], [12], [16], [13]. CNN architectures such as AlexNet [38], VGG [9], ResNet [39], GoogleNet [40], Inception Res-NetV2 [41], *etc.* are concerned primarily with improving accuracy. However, deploying these neural networks on edge devices is challenging due to their memory requirements, computational complexity, and power requirements. Optimization of the above mentioned parameters makes the architectures suitable for real-time applications. Several researchers focused on the development of efficient CNN architectures for image classification. Iandola et al. [37] proposed SqueezeNet, which is a lightweight CNN architecture for image classification comprising only 1.25 million parameters. It achieved comparable accuracy with AlexNet while utilizing 50 times fewer parameters [37]. In [12], MobileNet has been proposed for image classification as a lightweight CNN architecture for edge device applications. In a similar work, Sandler et al. [13], presents MobileNetV2, which offers better performance than MobileNet with 4 M parameters. Zoph et al. [16] developed a smaller version of NASNet, NASNetMobile, to address the needs of embedded and mobile computer vision applications. The proposed study presents Lite-MDC, a lightweight multi-kernel depth-wise convolutional network inspired by VGG16 and multi-kernel depthwise separable convolutions, which are suitable for real-world applications.

3. Datasets used

This section introduces a proposed novel pigeonpea dataset construction and various publicly available datasets used for comparison.

3.1. Pigeon-pea dataset construction

The scarcity of data limits the application of machine learning and deep learning-based methods in the field of agriculture. The inherent complexities associated with the environment and plant-related factors pose significant challenges in the creation of high-quality image dataset. To meet these challenges, the proposed work introduces the first novel pigeon-pea dataset. The flow of dataset construction is depicted in Fig. 2. The construction of the pigeon-pea dataset involves a three-step process: (1) System Establishment, (2) Dataset collection and annotation, (3) Dataset processing and verification.

3.1.1. System establishment

The work has been started by signing the research collaboration Memorandum of Understanding (MOU) between our Technical institute and Agriculture university. With the help of the Indian Council of Agricultural Research (ICAR) and the Indian Institute of Pulses Research (IIPR), twenty Agriculture Research Stations (ARS) located in eighteen states of India have been selected. To maintain a standard protocol for capturing the images from each ARS, we offer training programs to agriculture scientists for pigeon-pea disease image collection.

3.1.2. Data collection and annotation

To create a diverse pigeon-pea dataset, each research station collect images for four common pigeon-pea diseases: Wilt, Sterility mosaic, Dry root rot, Phytophthora blight, and healthy samples. The images are captured using several smartphone cameras and digital camera including samsung M21, Sony DSLR etc. Images are captured with multiple-resolution cameras, adding random spatial resolutions in the Dataset. All images are hand-labeled by agriculture scientists from the respective ARS. The statistics of the number of images from respective ARS and classwise distribution are given in Table 1 and Table 2 respectively. The dataset is curated from 18 distinct geographic locations as shown in Fig. 4, each characterized by varying backgrounds and environmental conditions. Images are taken at different times of the day with various scales of plant diseases, which enhances the dataset's overall robustness and adds uniqueness to its composition. Sample images with multiscale variation and background variation are shown in Fig. 3 and Fig. 5.

3.1.3. Dataset processing and verification

Following image collection, the dataset undergoes a preprocessing phase where undesired blurry images are eliminated. Furthermore, the images are uniformly cropped to a spatial resolution of 1024×1024 to ensure consistent spatial quality. Further each image is cross-verified by computer vision experts to avoid errors. All the images are resized using bilinear interpolation to $224 \times 224 \times 3$ before being applied to the CNN model. To standardize the data around the origin, data normalization involves the subtraction of the mean of each channel (RGB) from each pixel of the image. The proposed novel pigeon-pea dataset contains 12,232 images having five classes, including common pigeon-pea diseases. The proposed Pigeon-pea dataset offers extensive coverage and

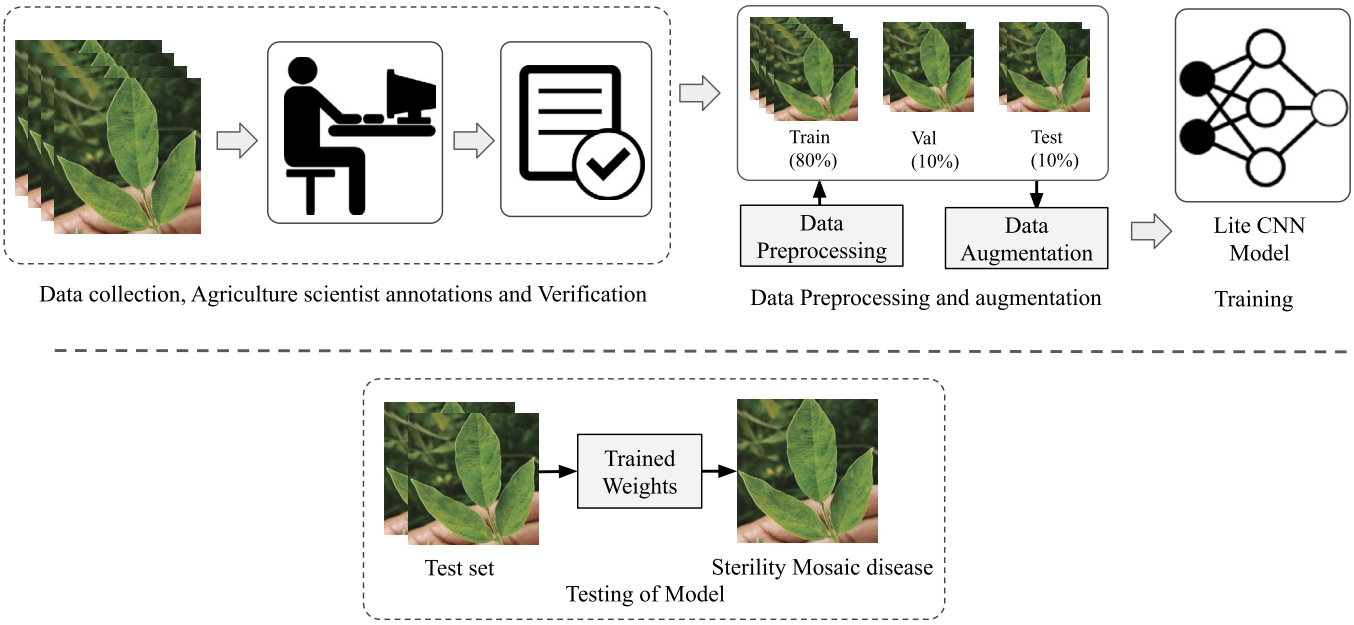


Fig. 2. Data collection, annotation by experts, data cleaning for novel pigeon-pea dataset.

Table 1
Data collection from Agriculture Research Stations (ARS) and Pulse Research Stations across India.

Sr. No.	Crop	Images collection location	No. of Images
1.	Pigeon-pea	Agriculture Research Station, Badnapur, Maharashtra	656
2.	Pigeon-pea	Agriculture Research Station, Ludhiyana, Punjab	600
3.	Pigeon-pea	Institute of Agricultural Science, Banaras Hindu University, Varanasi Uttar Pradesh	820
4.	Pigeon-pea	Agriculture Research Station, Bangaluru, Karnataka	745
5.	Pigeon-pea	Agriculture Research Station, Coimbatore, Tamil Nadu	786
6.	Pigeon-pea	Agriculture Research Station, Warangal, Telangana	623
7.	Pigeon-pea	Agriculture Research Station, Tirupati, Andhra Pradesh	513
8.	Pigeon-pea	Agriculture Research Station, Guntur, Andhra Pradesh	704
9.	Pigeon-pea	Agriculture Research Station, Dharwad, Karnataka	565
10.	Pigeon-pea	Agriculture Research Station, Kanpur, Uttar Pradesh	790
11.	Pigeon-pea	Agriculture Research Station, Khargone, Madhya Pradesh	889
12.	Pigeon-pea	Agriculture Research Station, Vijayapura , Karnataka	827
13.	Pigeon-pea	Agriculture Research Station, Junagadh, Gujarat	640
14.	Pigeon-pea	Pulses Research Station, Sardar Krushinagar , India	700
15.	Pigeon-pea	International Crops Research Institute for the Semi-arid Tropics, Patancheru, India	875
16.	Pigeon-pea	Agriculture Research Station Rahuri, India	587
17.	Pigeon-pea	Indian institute of Pulses Research, Kanpur, India	348
18.	Pigeon-pea	Pulses Research Station, Sardar Krushinagar , India	255
19.	Pigeon-pea	Pulses Research Station Junagadh, Gujarat, India	336
20.	Pigeon-pea	Birsa Agricultural University, Ranchi	369
		Total	12,232

Table 2
Class wise statistics for proposed Pigeon-pea Dataset.

Classes	Collection condition	Number of images	Resolution	Annotations type
Dry root rot	Real field images	2103	1024 x 1024	Image level
Healthy	Real field images	2560	1024 x 1024	Image level
Phytophthora blight	Real field images	1691	1024 x 1024	Image level
Sterility Mosaic	Real field images	3652	1024 x 1024	Image level
Wilt	Real field images	2226	1024 x 1024	Image level
	Total	12,232		

diversity of pigeon-pea diseases and is publicly available for research purposes on GitHub link[1]

[1] https://github.com/sandesh001/First-Pigeon-pea-datset-for-diseases-detection.

3.2. Publicly available datasets

The existing studies on plant diseases identification referred to the various datasets where detailed statics are mentioned in the Table 3. Amongst these various datasets three datasets are available publicly: (1) Cassava Dataset [42], (2) the Plant village dataset [6], and (3) the Apple

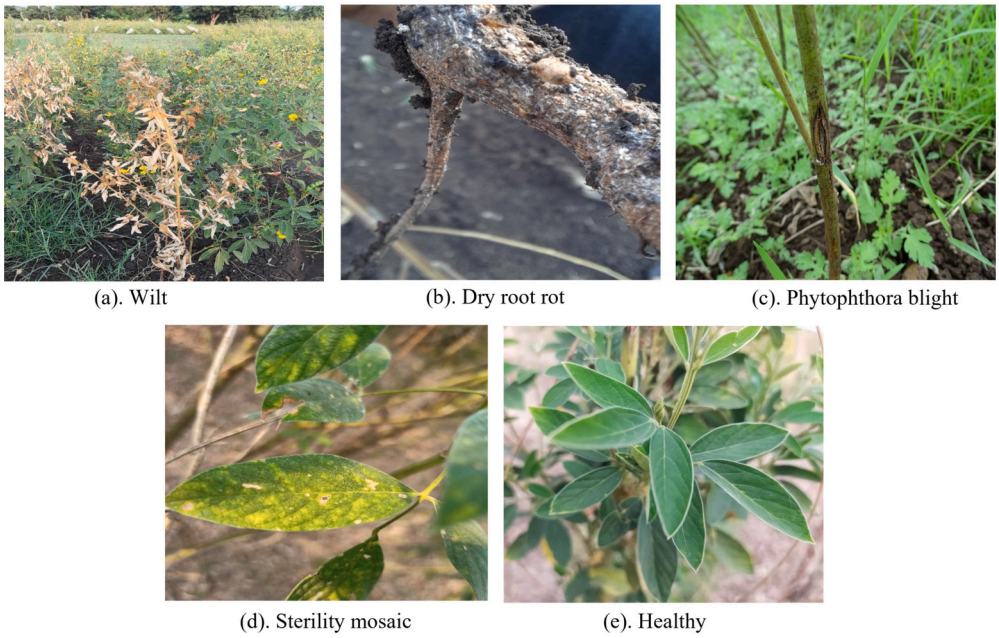


Fig. 3. Class wise sample images from the proposed Pigeon-pea dataset.

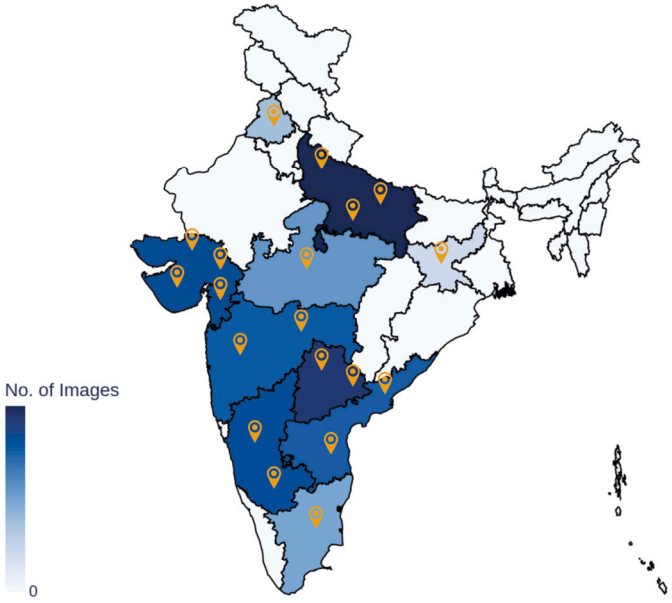


Fig. 4. Data collection from twenty geographic locations Agriculture Research Stations (ARS) and Pulse Research Stations across India.

dataset [43]. The proposed architecture is assessed on a total of four datasets, comprising the newly introduced pigeon-pea dataset, along with three publicly accessible datasets. The details of various datasets used in our work are given in Table 4. Fig. 6 showcase sample diseased images and healthy images from each dataset.

3.2.1. Cassava dataset

The cassava dataset [42] is the benchmark dataset for cassava disease detection. Around two hundred farmers contributed to the data from farmlands in different parts of Uganda. All images are captured in the entire field using smartphone cameras. This dataset consists of RGB images with five classes, including healthy plant leaves and diseased plant leaves representing the four diseases; Cassava green mite (CGM), mosaic disease (CMD), bacterial blight (CBB), and brown streak disease (CBSD). Sample images from the dataset are shown in Fig. 6

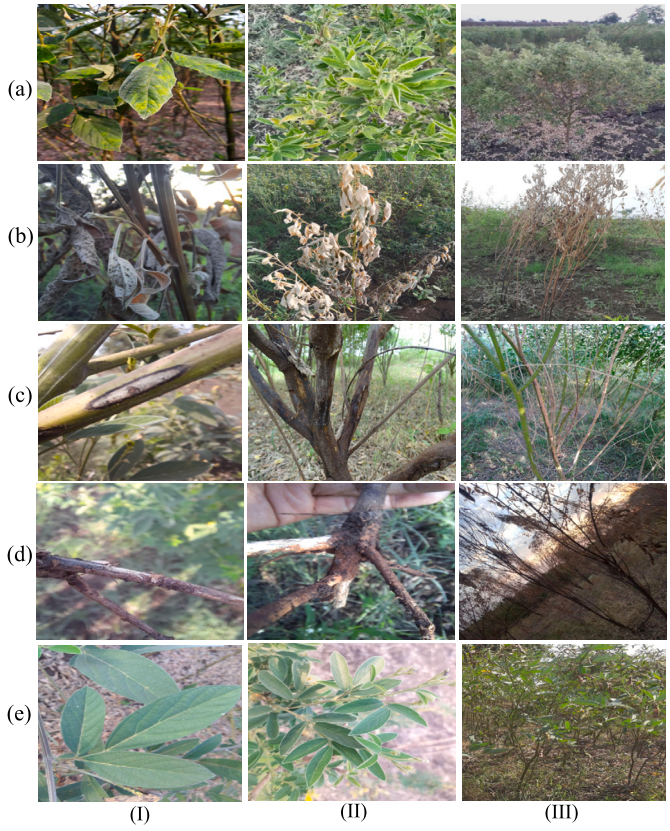


Fig. 5. Sample images from Pigeon-pea dataset (a) Sterility mosaic, (b) Wilt, (c) Phytophthora blight, (d) Dry root rot, (e) Healthy and Multi scale variation as shown in column (I) Closed, (II) Medium and (II) Far.

(b). There are 9,436 images labeled by agriculture experts with resolution of 4000 × 2672. The details mentioned in Table 4. The Cassava dataset is obtained from the "Cassava leaf disease competition" on the Kaggle platform. Since the ground truth for the test dataset is not available, the training dataset is divided into training, validation, and test subsets, with sizes of 60%, 20%, and 20%, respectively.

Table 3
Statistics of existing plant diseases dataset.

Dataset	Number of Images	Class Number	Collection condition	Diseases on	Status
Leaflet Cassava Dataset [44]	1,896	6	Field images	Only Leaf	Not available
Kaggle Cassava Dataset [42]	9,436	5	Field images	Only leaf	Publicly available
Leaf Diseases Dataset [30]	4,483	15	Field images	Only Leaf	Not available
Plant Village Dataset [6]	54,309	38	Lab Images	Only Leaf	Publicly available
Apple Foliar Disease Dataset [43]	3,651	4	Field images	Only Leaf	Publicly available
Apple Leaf Disease [45]	404	3	Field images	Only Leaf	Not available
Crop Pest Dataset [46]	4,483	40	Field images	Only Leaf	Not available
PDD271 [47]	220,592	271	Field images	Only Leaf	Not available
Pigeon-pea Dataset (Proposed)	**12,232**	**5**	**Field images**	**Entire Plant**	**Publicly available**

Table 4
Details of datasets used for experimentation.

Dataset	Collection condition	Resolution	Total Images	Classes
Kaggle Cassava Dataset [42]	Real field images	4000 × 2672	9,436	5
Apple Foliar Disease Dataset [43]	Real field images	1048 × 1365	3,651	4
PlantVillage Dataset [6]	Indoor controlled environment	256 × 256	54,309	38
Pigeon-pea Dataset (proposed)	Real field images	1024 × 1024	12,323	5
Total			94,700	



Fig. 6. Sample images from (a) Pigeon-pea dataset, (b) Kaggle cassava dataset, (c) Plane Village dataset and (d) Apple disease dataset.

3.2.2. PlantVillage dataset

The PlantVillage dataset [6] is a widely employed benchmark dataset within the domain of plant disease detection. It encompasses a total of 54,309 RGB images, each boasting a resolution of 256 × 256 pixels details can be seen in Table 4. These images are representative of 38 distinct plant disease categories, all are acquired under controlled indoor environmental conditions with a uniform background. A visual representation of the dataset can be seen in Fig. 6(c). Furthermore, for research and evaluation purposes, the dataset has been segmented into training, validation, and test subsets, comprising 60%, 20%, and 20% of the data, respectively.

3.2.3. Apple leaf dataset

The Apple leaf dataset [43] comprises multiple images of apple foliar diseases, which exhibit a range of lighting, angles, surfaces, and noise levels. These images are captured in real-world field conditions, using a Canon Rebel T5i DSLR and smartphones. The dataset covers total 3,651 RGB images with resolution of 1048 × 1365 pixels. These images are classified into four classes: apple scab, cedar apple rust, multiple diseases, and healthy leaves. Further details regarding the dataset are provided in Table 4, while sample images are shown in Fig. 6 (d). The Apple leaf dataset is obtained from the Plant Pathology Challenge competition on the Kaggle platform. Since the ground truth for the test dataset is not available publicly, the training dataset is split into training, validation, and test subsets, with sizes of 60%, 20%, and 20%, respectively.

4. Proposed methodology

The proposed method aims to design a lightweight network for plant disease detection. The primary objective is to create an efficient CNN architecture that is appropriate for real-time applications and can be deployed on resource constrained devices. Characteristics of efficient CNN architectures are (a) Less computational complexity (Total multiplication and addition operations), (b) Less number of parameters (c) Low memory requirement (model size) and (d) enhanced performance.

4.1. Convolutions in deep learning

Standard convolution is a core building block in deep learning models, notably for image and video analysis tasks. However, its computa-

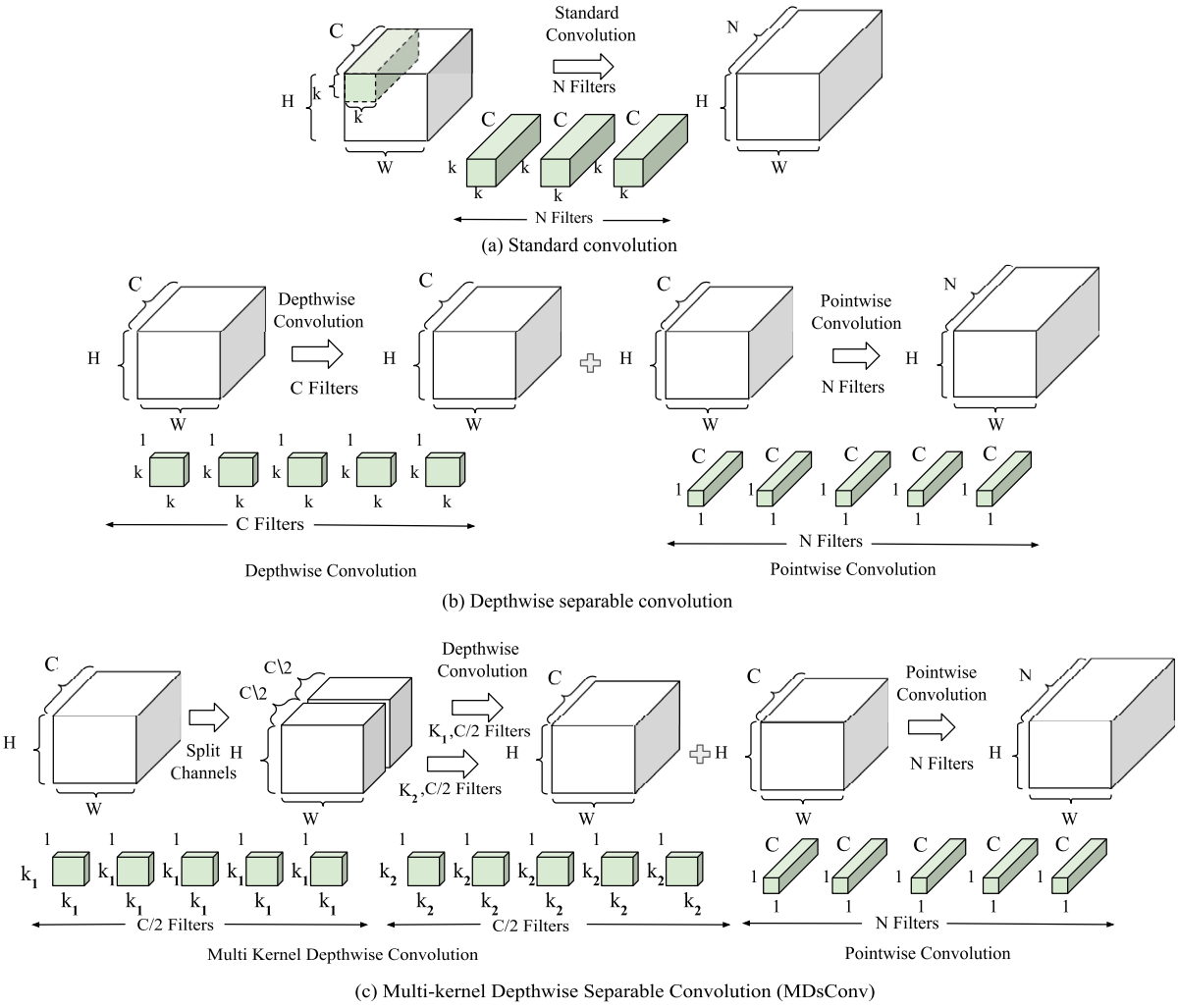


Fig. 7. Convolution operations (a) Standard convolution, (b) Depthwise convolution, and (c) Multi kernel Depthwise separable convolution (MDsConv).

tional demands can restrict its real-time applicability. The equations for computational cost and parameters for standard convolution are given in Eq. (1) and depicted in Fig. 7 (a). To address this limitation, depthwise convolution was introduced, which independently processes input channels with separate kernels, reducing parameters and computational cost. The combination of depthwise and pointwise (1x1) convolutions in depthwise separable convolutions (DWSC) has become a popular choice for building more efficient deep neural networks. The computational cost and parameters for DWSC are given in Eq. (2) and depicted in Fig. 7 (b).

$$Cost_Std = K \times K \times C \times N \times H \times W$$
$$Parm_Std = K \times K \times C \times N + N \quad (1)$$

Where, $K \times K$ = Kernel size, C = Number of input channels, N = Number of output channels and $H \times W$ = Spatial Height and Width of input image or feature map (For simplicity we consider padding = same, and stride = 0 and $H = W$).

$$Cost_DWsConv = Cost_DW + Cost_PW$$
$$Cost_DWsConv = (K \times K \times C_{in} \times H \times W) + (C_{in} \times C_{out} \times H \times W)$$
$$Parm_DWsConv = (K \times K \times C_{in} + C_{in}) + (C_{in} \times N + N) \quad (2)$$

4.2. Proposed multi kernel depthwise separable convolution

Depthwise separable convolution, while efficient, has limitations for multi-scale feature extraction. To over come this limitation we proposed the Multi kernel Depthwise separable Convolution (MDsConv) which uses multiple kernels in depthwise convolution operation to extract multiscale features of plant diseases. Thus multi-scale depthwise convolution helps to learn spatial correction at different scales. The input channels are grouped, where each group size can be calculated as per Equation (3).

$$G_s = \frac{C_n}{K_i} \quad (3)$$

Where, G_s = Group size, C_n = Number of input channels and K_i = Number of unique kernels. When $G_s = 1$ MDConv is equivalent to standard depthwise convolution. Multiple kernels with depthwise convolution operations are applied to each channel group. The proposed MDsConv includes 3×3 and 5×5 kernels, thus group size is $G_s = 2$, and input channels are divided into two groups. The number of 3×3 and 5×5 kernels is exactly half the number of channel in input feature map. The input feature map is partitioned into two groups as $I = [\frac{C}{2}, \frac{C}{2}]$, where first $\frac{C}{2}$ feature map corresponding to 3×3 DwConv and second $\frac{C}{2}$ feature map corresponding to 5×5 DWsConv is depicted in Fig. 7(c). Further, Pointwise convolution is used to fuse separable channels and helps in capturing cross-channel dependencies. The complete MDsConv

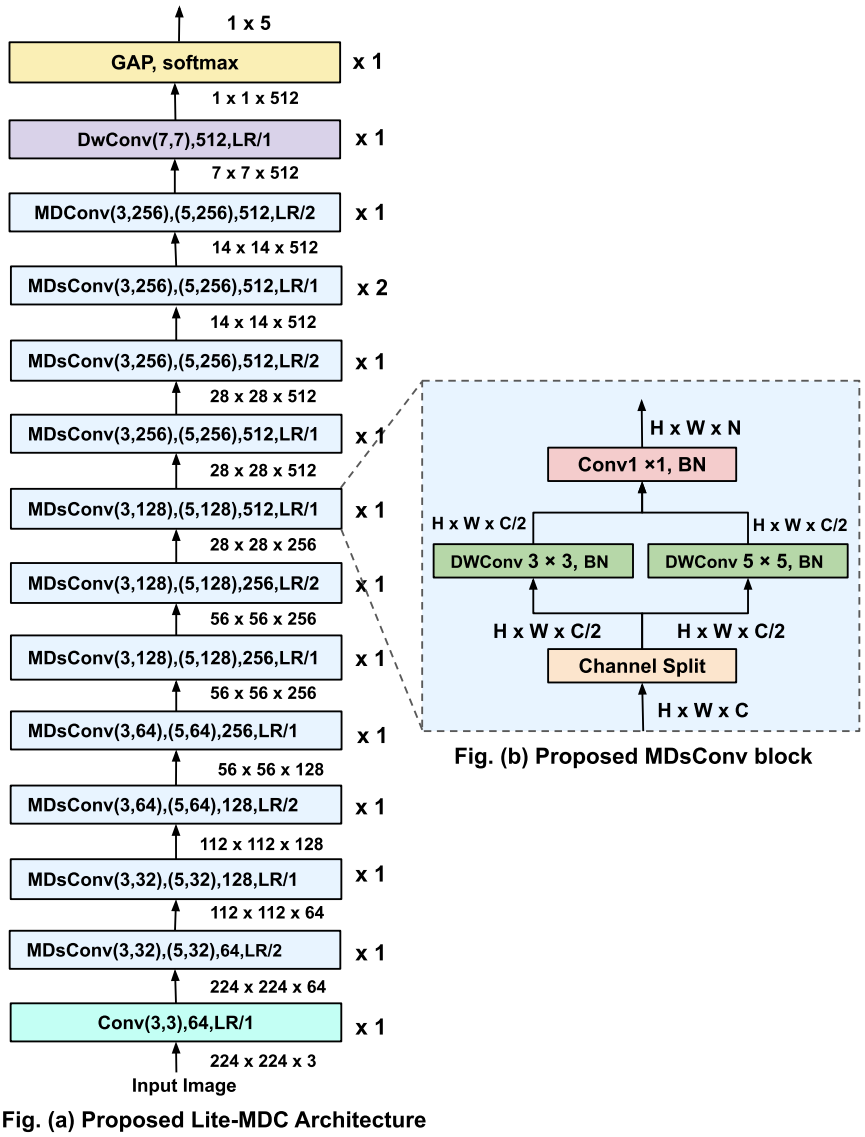


Fig. (a) Proposed Lite-MDC Architecture

Fig. (b) Proposed MDsConv block

Fig. 8. Fig. (a) shows proposed Lite-MDC architecture includes MDsConv blocks referred as MDsConv[(K1, No. of filters), (K2, No. of filters), No. of filters for pointwise convolution, LR/stride:LeakyReLU / 1 or 2] where, K1:3 x 3 kernel, K2:5 x 5 kernel, GAP: Global Average Pooling and Fig. (b) shows proposed MDsConv block where, DWConv: Depthwise convolution, BN: Batch Normalization, Conv: Standard convolution.

blocks shown in Fig. 7(c). Complete pseudocode for MDsConv is given in Algorithm 1. The overall computational cost and parameter required for MDsConv are computed as follows:

$$
\begin{aligned}
Cost_MDsConv &= Cost_MDConv + Cost_PW \\
Cost_MDsConv &= \left\{ (3\times 3\times \frac{C}{2}\times H\times W) + (5\times 5\times \frac{C}{2}\times H\times W) \right\} \\
&\quad + (C\times N\times H\times W) \\
Parm_MDsConv &= \left\{ (3\times 3\times \frac{C}{2} + \frac{C}{2}) + (5\times 5\times \frac{C}{2} + \frac{C}{2}) \right\} \\
&\quad + (C\times N + N)
\end{aligned}
\tag{4}
$$

4.3. Proposed lite-MDC architecture

The proposed Lite-MDC architecture uses a standard VGG16 network as a baseline architecture for plant disease classification. To overcome the limitations of standard VGG16 architecture, all the normal convolutions layers have been replaced with Multi kernel Depthwise separable Convolution (MDsConv). The proposed Lite-MDC architecture is depicted in Fig. 8. It takes input images with spatial dimensions

Algorithm 1: Multi kernel Depthwise separable Convolution (MDsConv).

Input: Input tensor (X_{in}), Kernel sizes [K_1, K_2]
Output: Output tensor X_{out}

```
G = len(X_in)
Kernels = [K_1, K_2]
Split = split(X_in, G, axis = −1)
Out = [ ]
# Multiscale depthwise convolution
for X_i, K_i in zip(Split, Kernels) do
    MDConv = DepthwiseConv2D(kernel_size = K_i)(X_i)
    Out.append(MDConv)
Out = Concatenate(Out)
Out = BatchNorm2D(Out)
# Pointwise convolution on concatenated output
Out = Conv2D(Out, kernel_size = 1)
X_out = BatchNorm2D(Out)
```

Table 5
Proposed lite-MDC architecture.

Layer Name	Filters size	Stride	Output shape
Input Image	-	-	224 × 224 × 3
Conv0	3 ×3 × 3× 64	1	224 × 224 × 64
MDConv_1	3 ×3× 32, 5 ×5× 32	2	112 × 112 × 64
PConv_1	1 ×1 × 64× 64	-	112 × 112 × 64
MDConv_2	3 ×3× 32, 5 ×5× 32	1	112 × 112 × 64
PConv_2	1 ×1 × 128× 128	-	112 × 112 × 128
MDConv_3	3 ×3× 64, 5 ×5× 64	2	56 × 56 × 128
PConv_3	1 ×1 × 128× 128	-	56 × 56 × 128
MDConv_4	3 ×3× 64, 5 ×5× 64	1	56 × 56 × 128
PConv_4	1 ×1 × 128× 256	-	56 × 56 × 256
MDConv_5	3 ×3× 128, 5 ×5× 128	1	56 × 56 × 256
PConv_5	1 ×1 × 256× 256	-	56 × 56 × 256
MDConv_6	3 ×3× 128, 5 ×5× 128	2	28 × 28 × 256
PConv_6	1 ×1 × 256× 256	-	28 × 28 × 256
MDConv_7	3 ×3× 128, 5 ×5× 128	1	28 × 28 × 256
PConv_7	1 ×1 × 256× 512	-	28 × 28 × 512
MDConv_8	3 ×3× 256, 5 ×5× 256	1	28 × 28 × 512
PConv_8	1 ×1 × 512× 512	-	28 × 28 × 512
MDConv_9	3 ×3× 256, 5 ×5× 256	2	14 × 14 × 512
PConv_9	1 ×1 × 512× 512	-	14 × 14 × 512
MDConv_10	3 ×3× 256, 5 ×5× 256	1	14 × 14 × 512
PConv_10	1 ×1 × 512× 512	-	14 × 14 × 512
MDConv_11	3 ×3× 256, 5 ×5× 256	1	14 × 14 × 512
PConv_11	1 ×1 × 512× 512	-	14 × 14 × 512
MDConv_12	3 ×3× 256, 5 ×5× 256	2	7 × 7 × 512
PConv_12	1 ×1 × 512× 512	-	7 × 7 × 512
DwConv_13	7 ×7× 512	1	1 × 1 × 512
PConv_13	1 ×1 × 512× 512	-	1 × 1 × 512
Conv_14	1 ×1 × 512× 512	1	1 × 1 × 512
Global Average Pooling	1 × 1		512
Softmax	Classifier		5

$224 \times 224 \times 3$ and passes them to further layer for feature extraction. All the layers of the Lite-MDC architecture are followed by batch normalization, where the activations scale the input layer. This aids in the improvement of accuracy and stability. The proposed network applies LeakyReLU activation function after each batch normalization layer except the last one in the network. The LeakyRelu is used to overcome the problem of dead neurons existing in Relu activation by introducing a slight slope in the negative region, enabling faster convergence and improving efficiency. A stride of two is utilized to reduce the spatial dimension of the input feature, in place of max pooling. Further, the fully connected layers are substituted with 1×1 pointwise convolution that has 512 filters to decrease the number of parameters without compromising accuracy. Finally, global average pooling layers are followed by the softmax layer produces class label output. The detailed architecture is shown in Table 5. Dropout avoids overfitting in the training process by randomly ignoring some of the neurons [48]. The increasing dropout rate ranges from 0.05 to 0.4 in the last few layers of the network, as depicted in Fig. 8.

4.4. Computational complexity calculations for lite-MDC

The number of parameters and computations (multiplications and addition) for image processing through complete network is calculated as follows: The color RGB images $224 \times 224 \times 3$ is given as input first convolution layer is $H \times W \times C$ The kernels size $K \times K$ is 3×3 and the number of kernels are $N = 64$. The total multiplication, addition and parameter that required for this layer are given in Eq. (5), Eq. (6), Eq. (7)

$$\begin{aligned} Mul_Conv0 &= H \times W \times C \times K \times K \times N \\ &= 224 \times 224 \times 3 \times 3 \times 3 \times 64 \end{aligned} \tag{5}$$

$$\begin{aligned} Add_Conv0 &= H \times W \times (K \times K \times C - 1) \times N \\ &= 224 \times 224 \times (3 \times 3 \times 3 - 1) \times 64 \end{aligned} \tag{6}$$

$$\begin{aligned} Parm_Conv0 &= C \times N + N \\ &= 3 \times 64 + 64 \end{aligned} \tag{7}$$

In the next layer multi kernel depth-wise convolution is carried out on the $(224 \times 224 \times 64)$ output of the previous layer (Conv0). We use two filters $K_1 \times K_1$ with size 3×3 and $K_2 \times K_2$ with size 5×5 there are $N = 64$ such kernels, stride = 2. The total number of multiplications, additions and parameter are given in Eq. (8), Eq. (9), Eq. (10)

$$\begin{aligned} Mul_MDWConv1 &= H \times W \times \frac{C}{2} \times (K_1 \times K_1 + K_2 \times K_2) \\ &= 112 \times 112 \times 32 \times (3 \times 3 + 5 \times 5) \end{aligned} \tag{8}$$

$$\begin{aligned} Add_MDWConv1 &= H \times W \times \frac{C}{2} \times \{(K_1 \times K_1 - 1) + (K_2 \times K_2 - 1)\} \\ &= 112 \times 112 \times 32 \times \{(3 \times 3 - 1) + (5 \times 5 - 1)\} \end{aligned} \tag{9}$$

$$\begin{aligned} Parm_MDWConv &= (3 \times 3 \times \frac{C}{2} + \frac{C}{2}) + (5 \times 5 \times \frac{C}{2} + \frac{C}{2}) \\ &= (3 \times 3 \times 32 + 32) + (3 \times 3 \times 32 + 32) \end{aligned} \tag{10}$$

Further, output of MDConv_1 $(112 \times 112 \times 64)$ is used to carry Pointwise convolution. Thus, a new feature map is created by linear combination of multi kernel depthwise convolutions output. The number of multiplications, additions and parameters used in the pointwise convolution are given in eq. (11), Eq. (12), Eq. (13)

$$\begin{aligned} Mul_PWConv1 &= H \times W \times C \times N \\ &= 112 \times 112 \times 64 \times 64 \end{aligned} \tag{11}$$

$$\begin{aligned} Add_PWConv1 &= H \times W \times (C - 1) \times N \\ &= 112 \times 112 \times (64 - 1) \times 64 \end{aligned} \tag{12}$$

$$\begin{aligned} Parm_PWConv1 &= C \times N + N \\ &= 3 \times 64 + 64 \end{aligned} \tag{13}$$

Similarly, the number of parameters and computational cost is calculated for each layer of the network. In total, there are 1167 M additions and multiplications and 2.2 M parameters for processing one image using the proposed Lite-MDC architecture.

5. Experimentation

This section discuses the Evaluation Metrics, model training and implementation details.

5.1. Evaluation metrics

The evaluation of the proposed model's performance includes metrics such as Precision, Recall, F1-Score, and Accuracy and are calculated using Eq. (14), (15), (17), and (16) respectively. Where, True positives (TP) denote correctly predicted classes, False Positive (FP) denotes predicted classes that are misidentified as positive, False Negative (FN) denotes predicted classes that are misidentified as negatives, and True Negative (TN) denotes correctly classified as negative classes.

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

$$Recall/Sensitivity = \frac{TP}{TP + FN} \tag{15}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{16}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{17}$$

5.2. Training and implementation

In this study, experiments are conducted on four datasets, namely the proposed pigeon-pea dataset and three publicly available datasets, namely Plant Village [6], Cassava [42], and Apple [43]. Training a deep neural network typically requires a substantial amount of data; data can be enlarged with augmentations techniques. The images are subjected to four different transformations, including rotation (range = 360), width shift (range = 0.2), height shift (range = 0.2), shear (range = 0.2), zoom (range = 0.2), brightness (range = 0.1-1.0), horizontal flip, vertical flip, and nearest fill mode. The random values are generated in the given range during run-time and the number of input images is increased by two folds. Data augmentation helps in the efficient training of the model and gives more generalization to the pigeon-pea diseases classifier.

The proposed Lite-MDC model is implemented using Keras with Tensorflow in the backend for training [49]. The training environment is Ubuntu 16.04 operating system, Intel 2.54 GHz Processor, 16 GB running memory (RAM), and NVIDIA Tesla P100 (GPU). The training process uses a batch size of 16, Adam optimizer [50], and categorical cross-entropy loss function. To reduce the learning rate for efficient model training, the validation loss is monitored, and if it remains constant for five successive epochs, the learning rate is reduced by a factor of 0.3. The end-to-end Lite-MDC model is trained for all four datasets with the same hyperparameters and environmental conditions for a fair comparison. Each Dataset is split into training, validation, and testing sets with a ratio of 60%, 20%, and 20%, respectively. The input image size is fixed at $224 \times 224 \times 3$. After training the model on the training dataset, its performance is evaluated on the test dataset using Precision, Recall, F1-Score, and Accuracy metrics.

6. Results and discussion

This section discusses the results and findings of the proposed approach and its evaluation on the proposed Dataset and three other publicly available datasets. The results are compared with state-of-the-art methods from both qualitative and quantitative perspectives to provide a comprehensive analysis. A lightweight CNN architecture with multi-kernel depthwise separable convolution called Lite-MDC is proposed for plant disease detection. Depthwise convolution is employed to reduce the computations, and the mixed convolution kernels are used to match the receptive fields in the multiscale variation of plant diseases. Detailed comparison of proposed MDsConv with Standard Convolution given as follows:

6.1. Comparison with standard convolution:

We compare MDsConv and StdConv's computational cost and parameter requirement for a single convolution operation. Cost is calculated as the sum of multiplication and addition operations. Let's consider the Input feature map $224 \times 224 \times 16$, input channels(C) = 16, Stride = 1, Padding = same, Number of filters (N) = 64. Standard convolution with Kernel size K = 3×3 cost and parameter comes out: Parameter = 9280, Cost = 921.6 M and with same setting cost and parameter for multi-kernel depthwise convolution kernel size $K_1 = 3 \times 3$, two kernels $K_2 = 5 \times 5$: Parameter = 1376, Cost = 14.20 M. MDsConv has seven times fewer parameters and 65 times less computational cost than standard convolution details shown in Table 6. The inference time is reduced by the decreased computations and parameters, resulting in fast performance. The proposed architecture has achieved state-of-the-art performance for plant disease classification while maintaining low computational complexity and memory requirements. Detailed computational complexity comparison is shown below:

Table 6
Comparison of parameters and cost calculation for single operation in **StdConv** and **MDsConv**, considering Input feature map shape = $(224 \times 224 \times 16)$, Stride = 1, Padding = 0, Output channels = 64. **StdConv** uses Kernel K = 3×3 and **MDsConv** uses two kernels $K_1 = 3 \times 3$, $K_2 = 5 \times 5$.

Method	Input	Parameter	Mul_Add	Kernels
StdConv	N = 64, C = 16	9,280	921.6 M	3×3
MDsConv	N = 64, C = 16	**1,376**	**14.20 M**	**3×3, 5×5**

6.2. Computational complexity comparison

As mentioned in section 4.4, Equations (5) to (13) are used to calculate the total computations and parameters utilized for the proposed Lite-MDC network. As a result, computations and parameters for all layers are 1167 M (additions and multiplications) and 2.2 M, respectively, for processing one image. The cost and parameters are significantly lower than required for a standard VGG16 network. The parameters and computational cost for VGG16 architecture are 15,300 M and 140 M receptively. A comparison of Lite-MDC and standard VGG16 architecture in terms of Parameters and computations are shown in Table 11. It is clear that Lite-MDC achieves significant improvement in accuracy with fewer parameters and computations, which further reduces the inference time and results in faster performance. Thus Proposed Lite-MDC Architecture is appropriate for Real-time application and outperformed the state-of-the-art methods on Pigeon-pea, Plant village, Cassava, and Apple datasets. The results of the experiments are presented in Tables 7 and 8.

6.3. Comparison with state-of-the-art methods on pigeon-pea dataset

The proposed architecture compares the performance of Lite-MDC with state-of-the-art methods like ResNet, Xception, InceptionV3, and DenseNet121 on the pigeon-pea dataset. In addition to the aforementioned networks, Lite-MDC is compared with lightweight state-of-the-art architectures like MobileNet, MobileNetV2, MobileNetV3, and NASNet mobile. The comparative details, such as accuracy, parameters, frames per second (FPS), and floating-point operations (FLOPS), are provided in Table 9. The proposed Lite-MDC method outperforms the existing methods and achieves 94.14% accuracy. The Lite-MDC uses only 2.2 million parameters, which is lesser among all networks mentioned in Table 9 and is 63 times less than the standard VGG-16. The accuracy versus several parameters plot for Lite-MDC and other state-of-the-art methods is given in Fig. 9. The fewer parameters in Lite-MDC are attributed to the multi-kernel depth-wise convolution. In addition, multiple kernels help to extract multiscale features of plant diseases. The obtained result shows that the method achieves the maximum accuracy with less inference time and the highest FPS. This demonstrates the effectiveness of the proposed Lite-MDC method and indicates its suitability for real-time plant disease detection. The classwise performance analysis with various metrics on the pigeon-pea dataset is mentioned in Table 8. The confusion matrix, Receiver operating characteristics (ROC), and area under the curve (AUC) are depicted in Fig. 10 (a) and Fig. 10 (b), respectively.

6.4. Comparison with existing studies on public datasets

The proposed method is evaluated on three publicly available datasets. (1) Plant Village dataset, (2) Cassava dataset, (3) Apple dataset. The comparison is carried out in terms of total parameters and accuracy, as shown in Table 10. The proposed architecture outperforms the existing work for all the datasets. All methods show good performance, around 99% on the plant village dataset Table 10, as all the images in this Dataset are captured in indoor conditions with a constant background and without multiscale variation samples shown in Fig. 6

Table 7
Comparison of proposed method Lite-MDC on different datasets.

Datasets	Precision	Recall	F1 Score	Accuracy (%)	Parameters
Cassava Dataset [42]	86	90	91	87.20%	2.2 M
Apple Foliar Disease Dataset [43]	86	90	91	97.36%	2.2 M
PlantVillage Dataset [6]	99	99	99	99.77%	2.2 M
Pigeon-pea Dataset (proposed)	94	94	95	95.01%	2.2 M

Table 8
Class-wise accuracy using proposed method architecture on Pigeon-pea dataset, Cassava and Apple leaf dataset.

Dataset	Classes	Total Samples	Right Predictions	Wrong Predictions	Precision	Recall	F1-Score	Accuracy (%)
Pigeon-pea (Proposed)	Dry root rot (DRR)	220	217	3	95	99	97	98.0
	Healthy	252	227	25	92	90	91	90.0
	Phytophthora blight (PB)	340	313	27	95	92	93.0	92.0
	Sterility mosaic (SM)	730	709	21	96	97	97.0	97.0
	wilt disease (WD)	445	417	28	95	95	95	93.7
Casava Leaf Dataset	Cassava Bacterial Blight (CBB)	109	76	33	56	50	52	50.0
	Cassava Brown Streak Disease (CBSD)	219	171	48	83	78	80	78.2
	Cassava Green Mottle (CGM)	210	183	27	84	77	80	77.0
	Cassava Mosaic Disease (CMD)	1280	1259	21	95	96	95	96.0
	Healthy	216	188	28	95	95	95	93.7
Apple Leaf Dataset	Healthy leaves	103	102	1	98	99	99	99.00
	Multiple diseases	18	11	7	83	56	67	61
	Apple Rust	221	124	97	96	99	98	99.0
	Apple Scab	119	117	2	97	98	98	98.3

Table 9
Comparison with state-of-the-art methods for pigeon-pea diseases detection.

Model	Approximate No. of Parameter	Accuracy	Frame per seconds (FPS)	FLOPS
Standard VGG16	140 M	87.33	5	31.1 G
ResNet50	25,63 M	79.34	7	7.85 G
InceptionV3	23.85 M	89.56	9	6.10 G
Xception	22.91 M	88.30	10	9.13 G
DenseNet121	7.0 M	86.37	8	7.04 G
EfficientNetB0	5.33 M	92.30	23	0.89 G
NASNet mobile	5.3 M	84.90	23	1.15 G
MobileNet	4.2 M	83.89	26	1.15 G
MobileNetV2	3.5 M	86.63	28	0.93 G
Lite-MDC (Proposed)	**2.2 M**	**94.14**	**33**	**0.61 G**

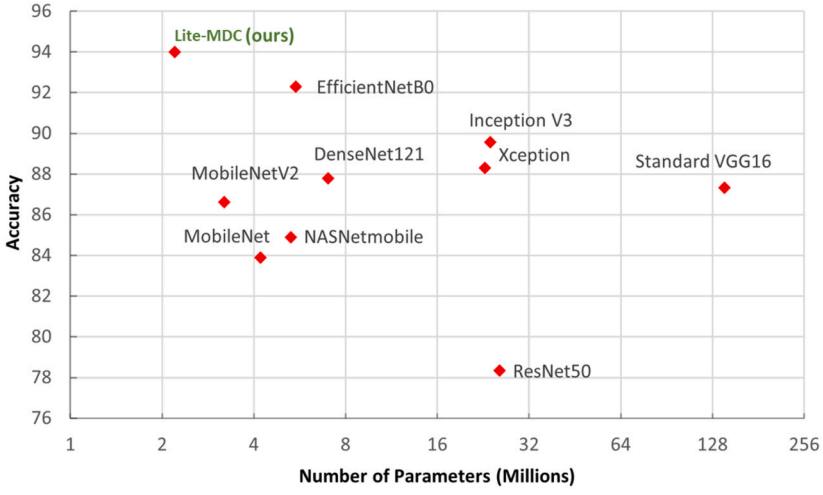
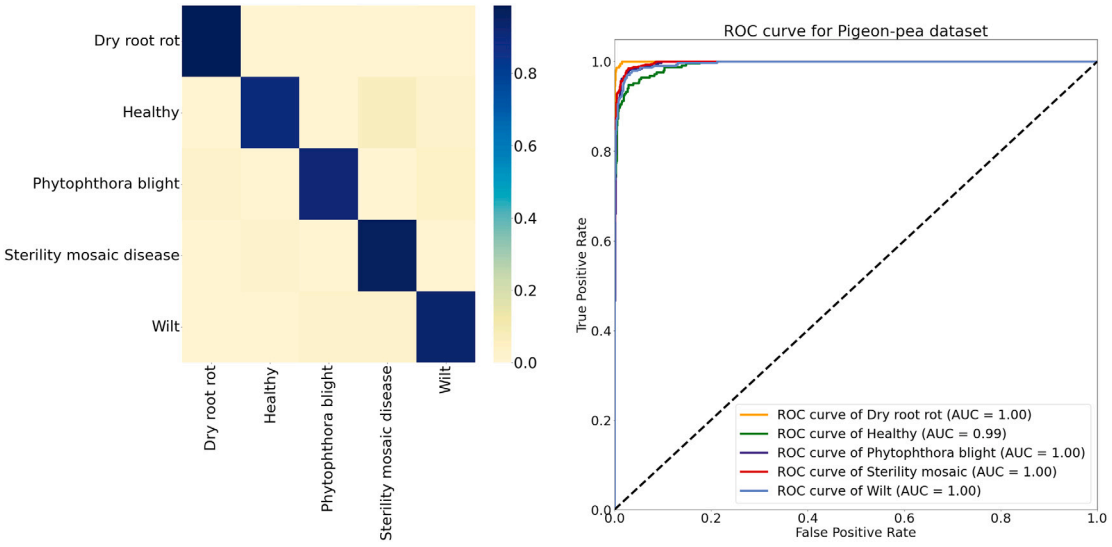


Fig. 9. Test Accuracy vs Number of Parameter on Pigeon-pea dataset, where x-axis shows Number of parameter in log scale and y-axis shows Test accuracy on pigeon-pea dataset.



(a) Confusion matrix for pigeon-pea dataset (b) ROC for Pigeon-pea dataset

Fig. 10. Proposed Lite-MDC (a) confusion matrix and (b) ROC Curve for Pigeon-pea dataset.

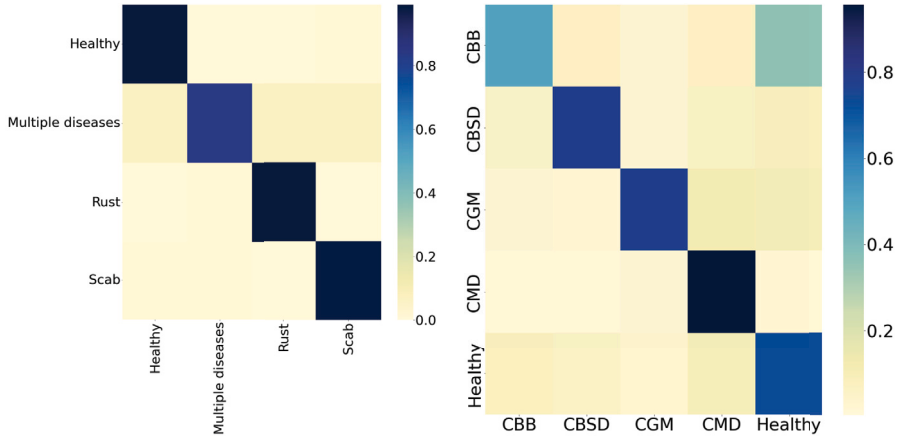
Table 10
Comparison of proposed Lite-MDC with state-of-the-art methods from the literature on Plan Village, Cassava, and Apple leaf Datasets.

Dataset	Method	Parameter	Accuracy %
PlantVillage [6]	AlexNet [38]	126 M	99.24%
	DenseNet169 [16]	14.31 M	99.89%
	GoogleNet [15]	23.85 M	99.76%
	ResNet34 [14]	21.6 M	99.67%
	ResNet152 [14]	60.42 M	99.69%
	VGG13 [9]	135.6 M	99.49%
	Squeezenet [37]	1.25 M	97.20%
	Xinda Liu et al. [47]	-	99.78%
	Lite-MDC (Proposed)	**2.2 M**	**99.78%**
Cassava Leaf Dataset [42]	VGG16 [9]	140 M	70.1%
	MobileNet [12]	4.25 M	68.5%
	ResNet50 [14]	25.64 M	81.5%
	H R Ayu et al. [51]	3.53 M	65.06%
	Alabdulmohsin et al. [52]	-	68.8%
	Lite-MDC (Proposed)	**2.2 M**	**86.4%**
Apple Leaf Dataset [43]	AlexNet [38]	62.37 M	65.6%
	VGG16 [9]	140 M	90.1%
	MobileNetV2 [12]	3.5 M	85.98%
	ResNetV2 [14]	44 M	93.68%
	InceptionV3 [15]	23.85 M	89.28%
	Manaswini et al. [53]	-	73.56%
	Trong et al. [54]	4.2 M	91.54%
	Alsayed et al. [55]	-	68.8%
	Lite-MDC (Proposed)	**2.2 M**	**97.2%**

(c); thus, it lacks in real-field scenarios. Further, to verify the effectiveness of the proposed model in actual field conditions proposed model is also evaluated on the Cassava and Apple leaf dataset. The images in the cassava dataset show the multiscale variation of cassava diseases. Table 10 shows that The Standard ResNet50 [14] model achieves 81.5% accuracy with 25.64 M parameters, whereas the proposed Lite-MDC model achieves 86.4% accuracy with 2.2 M parameters. The total parameters are reduced by eleven times as compared to ResNet50 architecture. The Apple leaf dataset has orchids images. Trong et al. [54] achieve 91.54% accuracy with 4.2 M parameters, whereas the proposed Lite-MDC model archives 97.2% accuracy with 2.2 M parameters. This shows the effectiveness of the proposed method for real-time disease detection. The classwise performance analysis of the proposed method is presented in Table 8 on Cassava, and Apple leaf datasets. The last four columns present evaluation metrics, Precision, Recall, F1-Score, and accuracy. The Table also shows total classwise samples and right and wrong predictions. The confusion matrix for the apple and cassava dataset is shown in Fig. 11 (a) and Fig. 11 (b) respectively. Receiver operating characteristics (ROC) and area under the curve (AUC) for the apple and cassava datasets are shown in Fig. 12 (a) and Fig. 12 (b), respectively. Fig. 9 shows the performance of the state-the-art method on all three datasets.

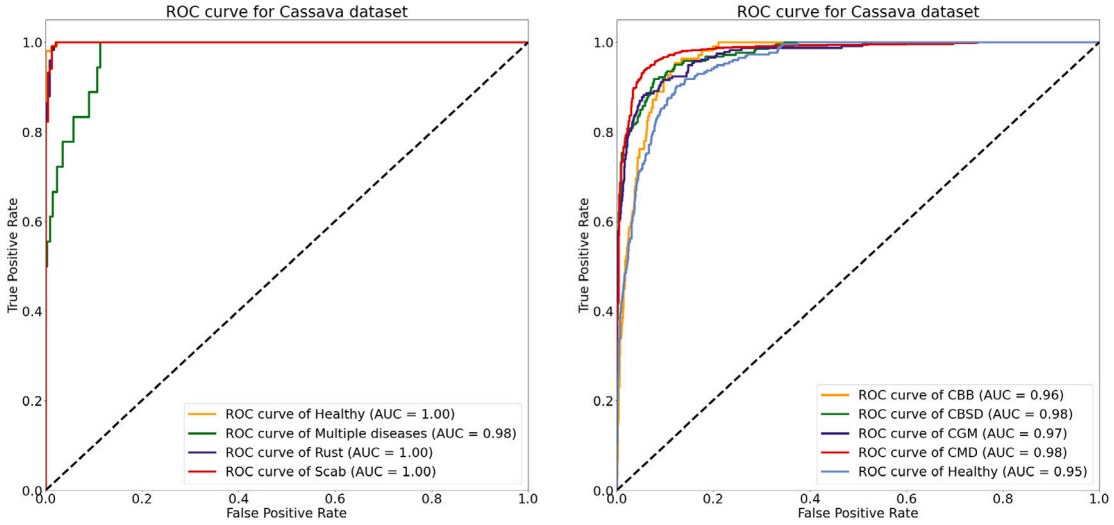
6.5. *Visualization:*

The gradient-weighted class activation heatmap is utilized to visualize attention of different methods on diseases part. [56] from the pigeon-pea dataset as depicted in Fig. 13 each row represents disease



(a) Apple dataset (b) Cassava dataset

Fig. 11. Confusion matrix for Apple and Cassava datasets in figure (a) apple dataset with four classes (b) cassava diseases as Cassava Bacterial Blight (CBB), Cassava Brown Streak Disease (CBSD), Cassava Green Mottle (CGM), Cassava Mosaic Disease (CMD) and Healthy.



(a) ROC for Apple dataset (b) ROC for Cassava dataset

Fig. 12. ROC-AUC curve for Apple and cassava dataset as shown in fig. (a) and fig. (b) respectively.

Table 11
Comparison of Lite-MDC and VGG16 in terms of parameters and computations.

Method	Computations	Accuracy	Total Parameter
VGG16 [9]	15,300 M	87.33%	140 M
Lite-MDC (Proposed)	**1,167 M**	**95.34%**	**2.2 M**

as (I) Sterility mosaic, (II) Healthy (III) Wilt, (IV) Dry root rot, (V) Phytophthora blight, and columns represent the various methods as Pro- posed Lite-MDC, DenseNet121, ResNet50, and MobileNetV1 in (a), (b), (c) and (d) respectively. Compared to the standard state-of-the-art methods, it is clear from visualization that the proposed Lite-MDC architecture can focus on diseased parts efficiently. The DenseNet121, ResNet50, and MobileNetV1 probably show a distracting focus on disease regions and ignore some useful disease features. Thus proposed multi-kernel approach can capture multi-scale features and thus correctly focuses on discriminative regions.

6.6. Ablation study

We are aiming to design lightweight architecture for real-time plant disease detection. Standard VGG16 was selected as the baseline architecture having state-of-the-art performance on benchmark imagenet datasets for image classification. The biggest problem with the standard VGG16 network is its huge computations and parameters. To address this problem, we identify layers in the network which require more computations and parameters and try to reduce parameters in the following steps: First, we found that fully connected layers in VGG16 are expensive in terms of computations and parameters. Therefore, we replace these layers with depthwise convolution while maintaining state-of-the-art performance, referred to as Modification-1. Second, further standard convolutions are replaced with depthwise convolutions to reduce parameters while maintaining state-of-the-art performance, referred to as Modification-2. From experiments, it is observed that depthwise convolutions with only one kernel 3×3 cannot extract multiscale features of plant diseases and have less accuracy. Finally, depthwise convolution is replaced with multi-kernel depthwise separable convolution (MDsConv) with kernel size 3×3 and 5×5 to address the problem of multiscale features of plant diseases referred to as Modification-3. Further, Dropout is used in lower layers of the Lite-MDC network with an increased dropout rate of (0.1 to 0.5) to avoid overfitting, referred to as Modification-4. The proposed Lite-MDC outperforms the existing state-of-the-art methods with 95% accuracy with 2.2 M parameters as mentioned in Table 12.

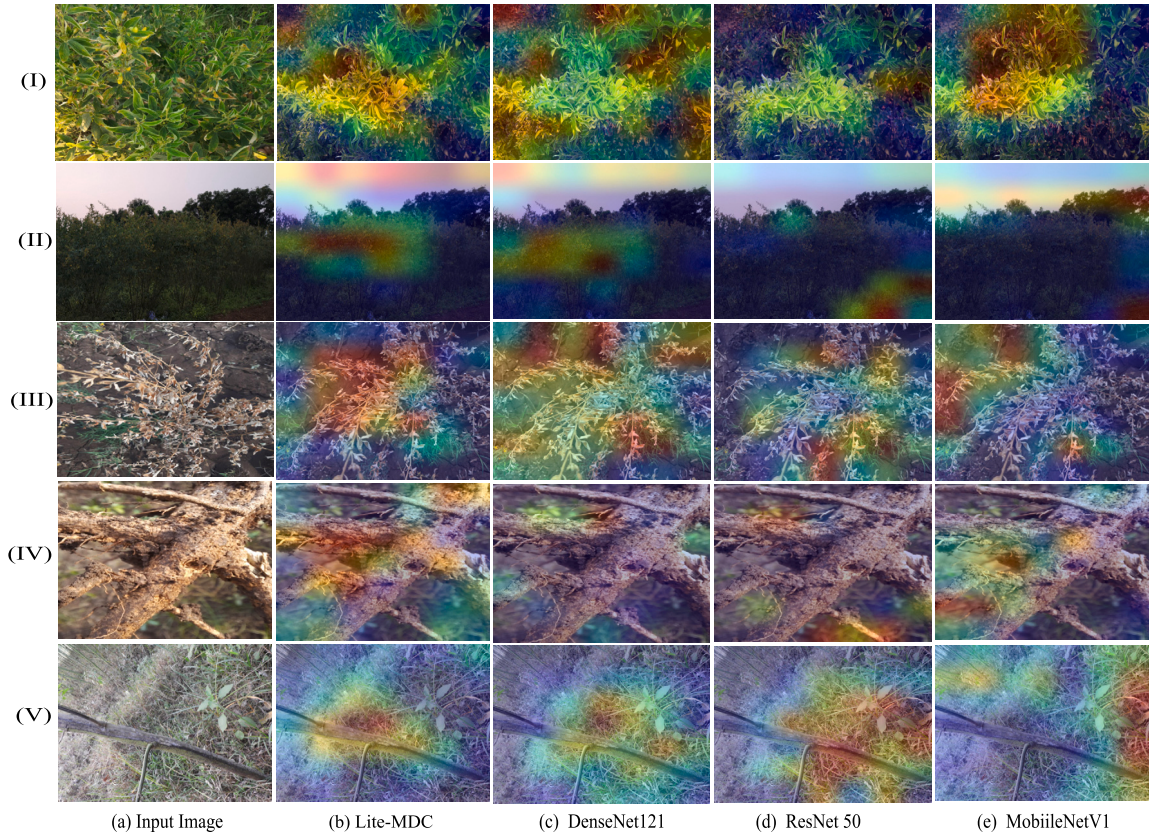


Fig. 13. Qualitative results GARD-CAM visualization comparison with other methods, row indicates disease classes and column indicates models (a) Input Image, The feature map from the last convolution layer of (b) Lite-MDC, (c) DenseNet121, (d) ResNet50, (e) MobileNet and diseases as (I) Sterility mosaic, (II) Healthy (III) Wilt, (IV) Dry root rot, (V) Phytophthora blight.

Table 12
Ablation study of Lite-MDC on Pigeon-pea Dataset (**DWConv**: Depthwise convolution, **MDsConv**: Multi Kernel Depthwise separable Convolution, **FC**: Fully Connected layer).

Methods	(DWConv in FC)	DWConv	MDsConv	Dropout	Accuracy	Parameter	FLOPS
VGG-16					87.33	140 M	31.0 G
Lite-MDC-Mod1	✓				90.53	15 M	10.70 G
Lite-MDC-Mod2	✓	✓			91.58	2.2 M	0.60 G
Lite-MDC-Mod3	✓		✓		93.80	2.21 M	0.61 G
Lite-MDC (Proposed)	✓		✓	✓	**94.14**	**2.21 M**	**0.61 G**

The suggested Lite-MDC architecture is well-suited for devices with limited resources, including agricultural robots and smartphones, etc. Mobile phones can serve as a convenient tool for farmers and experts to detect pigeon-pea diseases. Scouting robots are widely used for agriculture monitoring which helps farmers to optimizes agricultural practices. Moreover, the proposed architecture can be utilized in scouting robots for diseases detection and identification. However, it's essential to note that as a classification model, it does not pinpoint the exact disease location in an image, which constitutes a significant limitation of the proposed approach. Limitation can be addressed by using object detection architectures with bounding box annotations. It will be taken as a further research work in near future.

7. Conclusion

The first novel generalized pigeon-pea dataset collected from eighteen geographical locations for plant disease detection along with Lite-MDC method for optimized computations are presented in this work. This plant disease detection model takes 2.2 M parameters. The proposed architecture achieves high accuracy and efficient in real time computations and memory requirement. The multi-kernel depthwise convolution with 3×3 and 5×5 kernels improve the multiscale feature extraction for plant disease detection. The depthwise convolution helps to reduce computational complexity and network parameters. The proposed method uses sixty-three times less parameters compared to VGG16 architecture. Furthermore, the comparison of results between InceptionV3, VGG16, ResNet50, DenseNet, MobileNet, MobileNetV3, NASNet, SqueezeNet, and the proposed method demonstrates superiority in accuracy and speed, with 94.14% accuracy and a processing speed of 33 Frame Per Second (FPS) inference on NVIDIA P100 GPU with 16 GB RAM. The performance of the proposed method is validated on four datasets: the proposed novel pigeon-pea dataset, Plant Village dataset, Cassava dataset, and apple leaf dataset with 94.14%, 99.78%, 86.4%, and 97.2% accuracy, respectively. In future, the real-time performance of the proposed method can be validated on resource constrained devices such as smartphones, agriculture robots to impact real-time plant disease detection.

CRediT authorship contribution statement

Sandesh Bhagat: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project adminis-

tration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Manesh Kokare:** Supervision. **Praful Hambarde:** Supervision. **Trupti Taori:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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